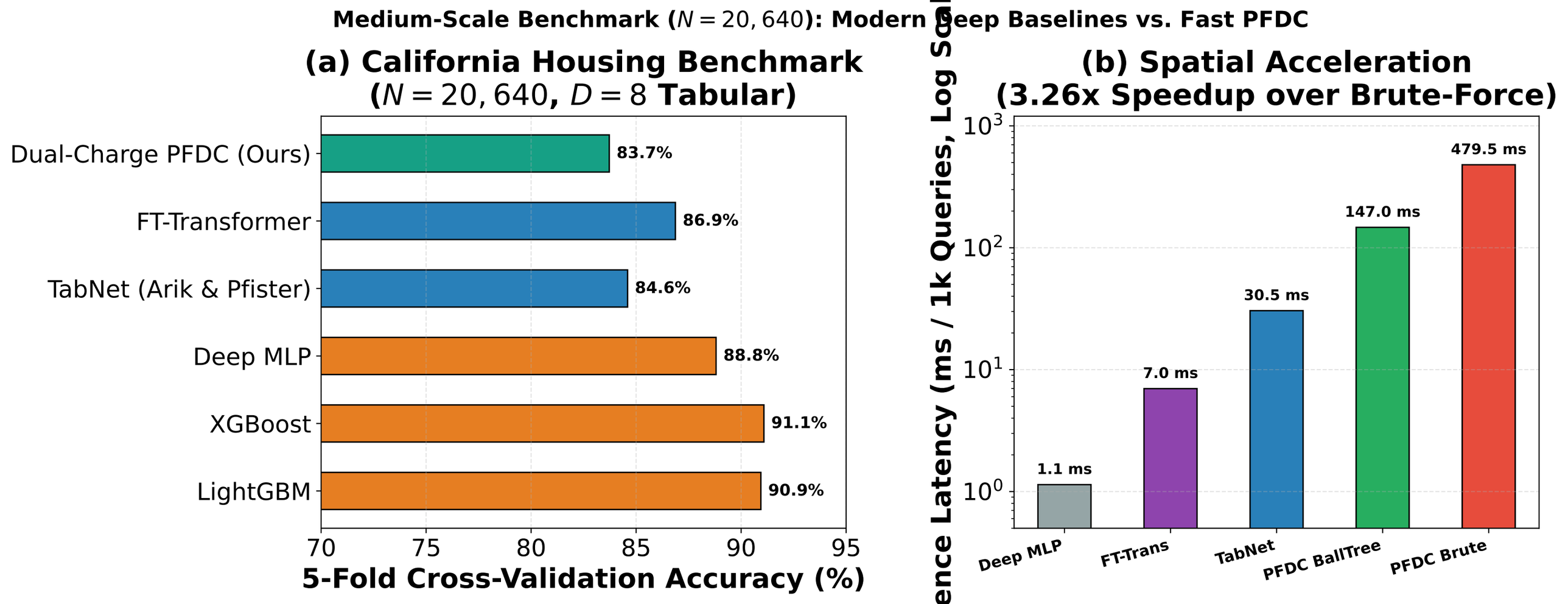
Medium-Scale Benchmark ($N = 20,640$): Modern Deep Baselines vs. Fast PFDC
(a) California Housing Benchmark
($N = 20,640$, $D = 8$ Tabular)
Dual-Charge PFDC (Ours)
FT-Transformer
TabNet (Arik & Pfister)
Deep MLP
XGBoost
LightGBM
83.7%
86.9%
84.6%
88.8%
91.1%
90.9%
70
75
80
85
90
95
5-Fold Cross-Validation Accuracy (%)
(b) Spatial Acceleration
(3.26x Speedup over Brute-Force)
ence Latency (ms / 1k Queries, Log Scale)
10^3
10^2
10^1
10^0
1.1 ms
7.0 ms
30.5 ms
147.0 ms
479.5 ms
Deep MLP
FT-Trans
TabNet
PFDC BallTree
PFDC Brute